Joosung (Benjamin) Park

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Research Interests: Human–AI Collaboration, Intent-Aware Agents, Multimodal Learning

EDUCATION

Penn State University — BS in Computer Science; Minors in Mathematics & Art | GPA: 3.43 Expected May 2027

EXPERIENCE

Undergraduate Research Assistant, LPAC, Penn State University October 2026 – Present

- Set up participants and collect data during human motion-capture sessions for lab studies

MC REU 2026, Penn State University — Advisor: Dr. Mahfuza Farooque June – August 2026

- Built a mutimodal AI system, which infers a Minecraft builder's goal from raw observation: frozen VPT, MineCLIP, and Uni3D encoders feed learned likelihood heads and a Bayesian belief over five goal categories
- Reached 0.375 goal accuracy on 24 unseen real sessions, the best of five methods tested (LLM scene reader 0.292, template baseline 0.042)
- Trained a 17.5M-parameter Transformer decoder that predicts the builder's next action token with 71% accuracy (59% eight tokens ahead); pretraining on unlabeled real sessions cut its loss 37%

MC REU 2025, Penn State University — Advisor: Dr. Shahid Hussain June – August 2025

- Built a Python pipeline that runs clang-tidy over 10 open-source C++ game engines and builds a 34,433-function dataset, split by engine so test engines are never seen in training
- Fine-tuned DeepSeek-Coder-1.3B LoRA adapters to detect clang-tidy findings and generate candidate repairs
- Reached 0.766 micro-F1 on 3,940 functions from two unseen engines (length baseline 0.133); 53.9% of 128 held-out repairs passed static re-analysis vs. 14.1% untuned

Penn State College of Engineering

Academic Integrity Student Committee — review integrity cases and policy July 2026 – Present

Front Desk Student Worker — schedule and check in students for advising August 2025 – Present

Undergraduate Research Ambassador (CEOI) — help students find faculty and research openings June 2025 – Present

Honor Guard, Staff Sergeant, Republic of Korea Air Force August 2022 – June 2024

- Performed ceremonial drill at 100+ public events, national ceremonies, and state functions

PROJECTS

PathFinder — Yale Hackathon 2026 March 2026

- Built the ML and backend of a satellite-image disaster-response tool with one teammate: YOLO damage detection, FastAPI routing, and geo-referencing

RippleJourney — ripple-journey.vercel.app March 2026

- Built a full-stack app in 3 days (Next.js, TypeScript, Prisma, PostgreSQL/pgvector) that helps students turn a research idea into a plan and find matching professors through Google Scholar

LEADERSHIP & ACTIVITIES

Mentor, Commonwealth Engineering Scholars Program — help transfer students from other Penn State campuses settle in and get started in research 2026 – Present

Team Member, Builder's Collective — hackathon team; shipped PathFinder at Yale 2026 2026 – Present

Member, Society of Hispanic Professional Engineers (SHPE) — Penn State University 2026 – Present

Member, Rowing Team (Crew) — train at daily 5:30 AM practices 2025 – Present

Representative, International Student Organization — mentored 10+ new international students 2021 – 2022

Team Member, League of Legends Esports — set strategy with a five-player roster; won ECAC 2021 – 2022

TECHNICAL SKILLS

Programming: Python, Java, JavaScript

Libraries & Frameworks: PyTorch, Hugging Face Transformers, PEFT, NumPy, SciPy, FastAPI, pytest, Claude CLI

AI/ML: LLM Fine-Tuning, LoRA, Computer Vision, Deep Learning, Transfer Learning, Multimodal Learning, Bayesian Inference, Transformer Models

3D Processing: Point Clouds, Voxel Representations, Geometric Template Matching

Engineering: Data Pipelines, Event-Driven Systems, Finite-State Machines, Software Testing, Git

Research: Experimental Design, Cross-Validation, Model Calibration, Baseline Comparisons